Socially Robust AI

Maria Teleki

PhD Candidate, Computer Science
Texas A&M University



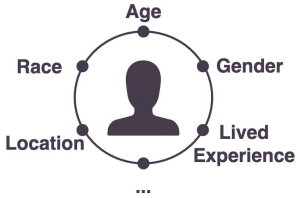
I build methods, benchmarks, and audits that **improve AI robustness in socially complex situations.**

My research spans Speech AI and Computational Social Science.

Maria Teleki

Socially Robust AI







Is AI robust to how people naturally speak?

TACL ‘26, ICASSP ‘26
INTERSPEECH ‘25/‘24,
LREC-COLING ‘24 x2

Is AI robust to each person’s uniqueness?

ICWSM ‘25, INTERSPEECH ‘26
ACL ‘26, EMNLP Findings ‘23/’25,
IUI ‘26, CHI ‘26

Is AI robust within institutions?

FAccT ‘26
L@S ‘26

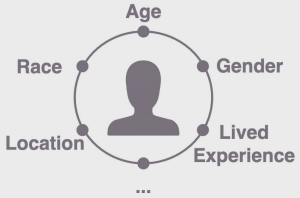
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Speech AI is everywhere



Voice Assistants

Siri Alexa Claude, ChatGPT, Gemini

all have Voice Mode

Healthcare
Transcription
Telehealth
Automated triage
Biomarkers for dementia, depression, ALS, and more
Check-in systems
Record keeping (SOAP)
Patient scheduling

Education
Online courses
Tutoring & coaching systems

Legal
Court transcription

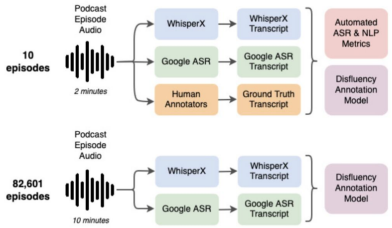
Entertainment
Dubbing
Music generation

Accessibility
Live translation (incl. code-switching)
Expressive captions
Illiterate users

Business
Meeting summarization, proactive agents
Voice-to-code programming
In-car voice assistants
Voice banking (finance)

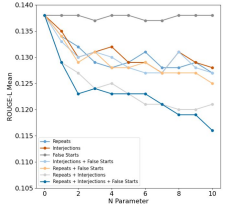
My research established...

Disfluencies are OOD

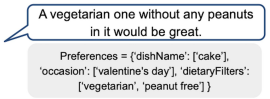


INTERSPEECH ‘24

Disfluencies harm task performance in...



summarization
LREC-COLING ‘24

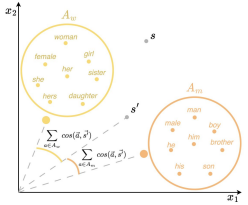


rec. systems.
INTERSPEECH ‘25



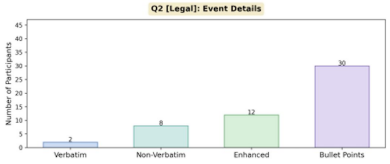
translation
TACL ‘26

Disfluencies carry identity information, e.g.,



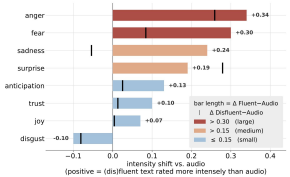
gender
ICWSM ‘25

Disfluency removal or preservation depends on task



In Prep for TOCHI

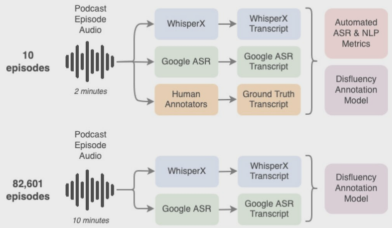
Disfluencies carry emotional information



TACL ‘26

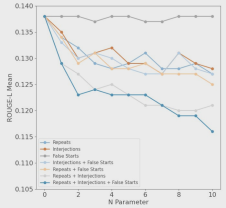
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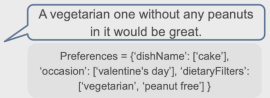


INTERSPEECH ‘24

Disfluencies harm task performance in...



summarization
LREC-COLING ‘24

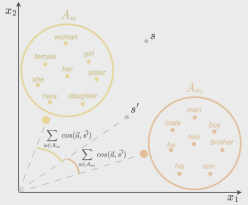


rec. systems.
INTERSPEECH ‘25



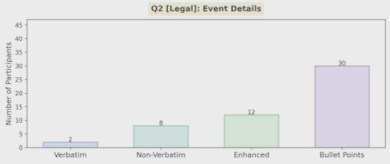
translation
TACL ‘26

Disfluencies carry identity information, e.g.,



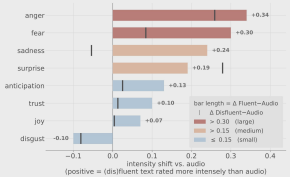
gender
ICWSM ‘25

Disfluency removal or preservation depends on task



In Prep for TOCHI

Disfluencies carry emotional information



TACL ‘26

Disfluency-Aware Translation



Applications in healthcare, business, disaster response, and more

meetween

The disfluencies could mean...

→ trying to remember what their dose is

→ uncertainty about the dose

→ anxiety about talking to a doctor (power imbalance)

→ searching for words in a second language

→ a speech or language condition (stuttering, aphasia)



How do speech translation systems translate disfluencies? *TACL '26*

Uh-Mazing benchmark: 80 Switchboard utterances x 8 target languages = 640 human translations with annotated disfluency spans

→ **Human study** showing disfluencies carry emotional information

→ **Taxonomy of failure modes**, including disfluency-specific failure modes

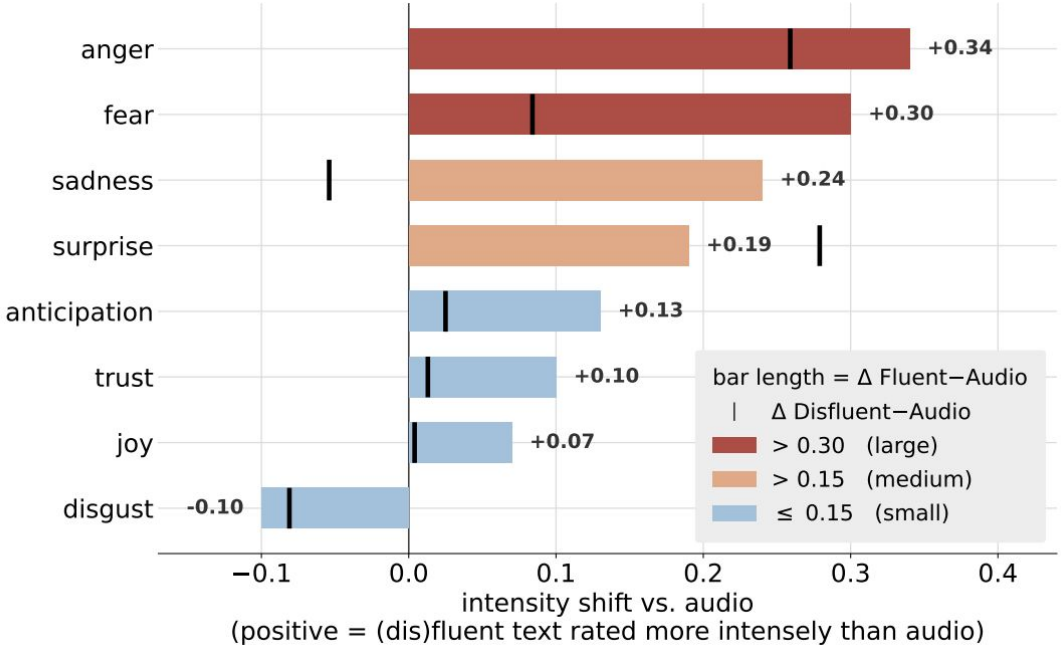
→ **Inference-time mitigation strategies** (beam search, in-context learning)



How do speech translation systems translate disfluencies? *TACL '26*

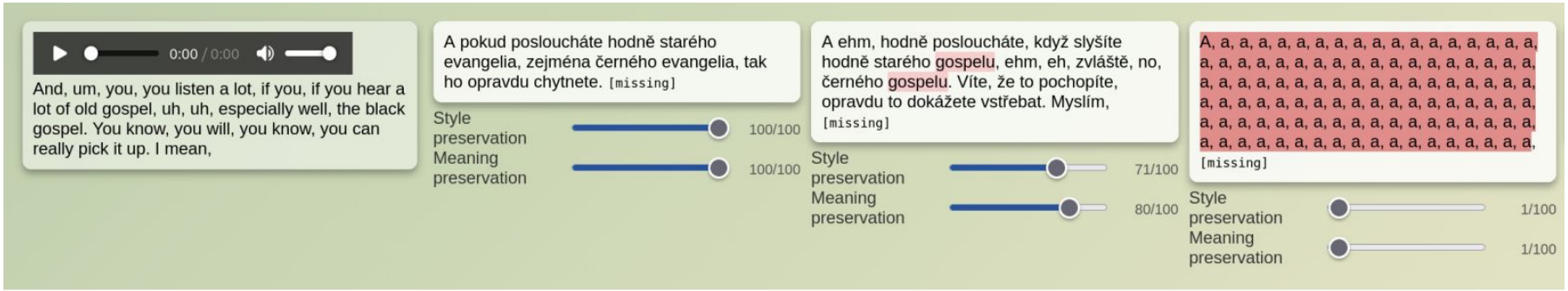
We conducted a human study showing disfluencies carry emotional information:

Key Finding:
Removing disfluencies **increases** the perceived **intensity of negative emotions**





How do speech translation systems translate disfluencies? *TACL '26*


0:00 / 0:00
And, um, you, you listen a lot, if you, if you hear a lot of old gospel, uh, uh, especially well, the black gospel. You know, you will, you know, you can really pick it up. I mean,
A pokud posloucháte hodně starého evangelia, zejména černého evangelia, tak ho opravdu chytnete. [missing]
Style preservation 100/100
Meaning preservation 100/100
A ehm, hodně posloucháte, když slyšíte hodně starého gospelu, ehm, eh, zvláště, no, černého gospelu. Víte, že to pochopíte, opravdu to dokážete vstřebat. Myslím, [missing]
Style preservation 71/100
Meaning preservation 80/100
A, a, a, a



How do speech translation systems translate disfluencies? *TACL '26*

		by language									by model				
		AR	CS	DE	ES	FR	HI	IT	ZH	total	canary	gemini	tower	phi	total
General Errors	Incorrect language ★	203	93	33	49	76	42	141	84	**721**	117	104	207	293	**721**
	Missing content ★	67	34	52	58	44	23	193	50	**521**	133	151	84	153	**521**
	Extra content	13	10	30	38	17	11	70	30	**219**	47	56	55	61	**219**
	Looping	19	23	6	6	4	16	1	7	**82**	2	4	15	61	**82**
Disfluency Errors	Deletion ★	11	68	144	208	184	88	238	157	**1098**	353	276	165	304	**1098**
	Extra disfluency	55	5	38	27	8	19	61	9	**222**	32	36	82	72	**222**
	Type mismatch	19	27	6	17	11	35	101	8	**224**	50	49	67	58	**224**
	Other error		1	2	2					**5**		3	2		**5**
Other Errors	Uncategorized error	9	3	17	10	20		1	2	**62**	14	12	17	19	**62**
		396	**264**	**328**	**415**	**364**	**234**	**806**	**347**	**3154**	**748**	**691**	**694**	**1021**	**3154**

We identify a **taxonomy of failure modes**.

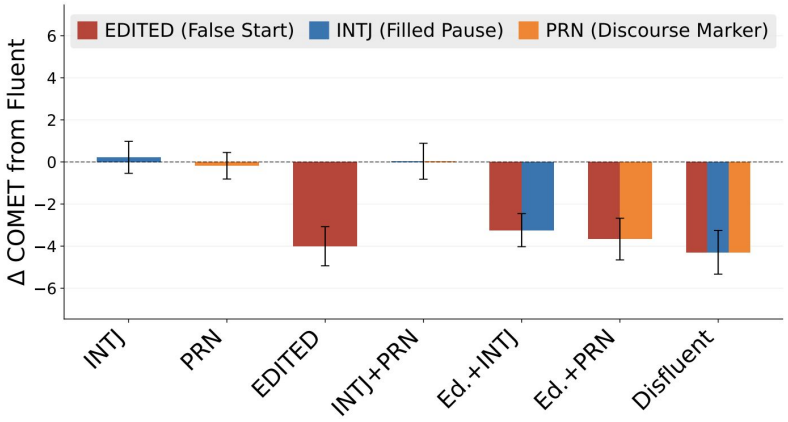


How do speech translation systems translate disfluencies? *TACL '26*

We make a **linguistic diagnosis** of what drives the failures:

EDITED disfluencies **drive the drop in performance**; some can be **recovered, detailed next...**



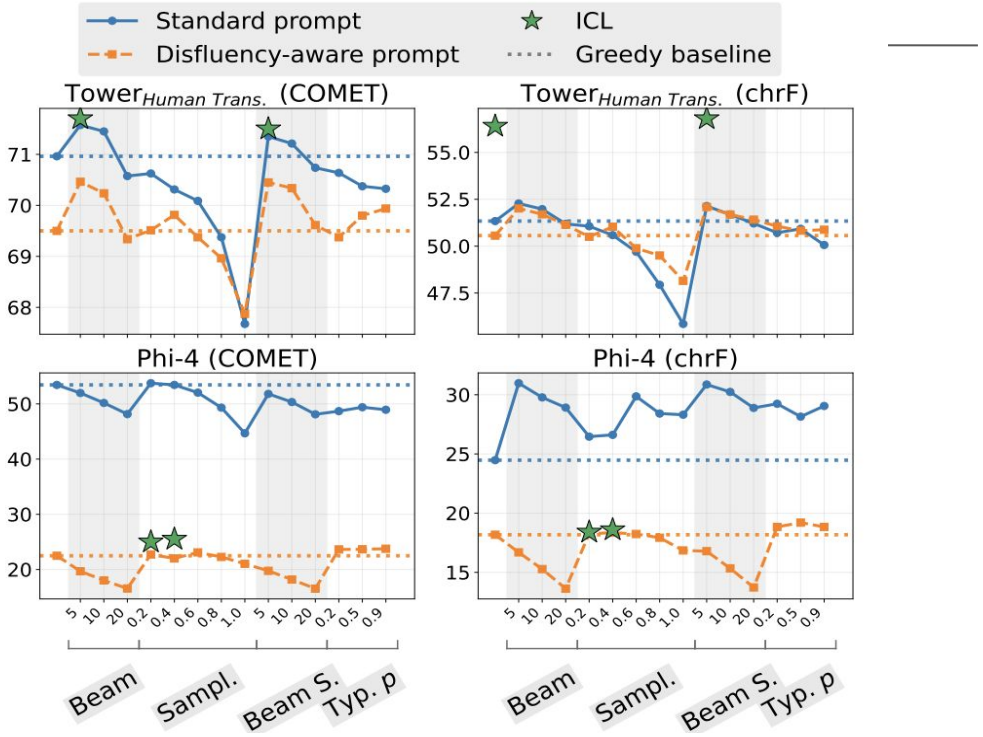




How do speech translation systems translate disfluencies? *TACL '26*

We contribute **inference-time mitigation strategies** (beam search, in-context learning) to recover some of the performance loss:

Decoding methods improve overall quality (COMET), while **ICL improves disfluency preservation** (chrF)



DeepMind

Current Work: “Well, actually –”: Social Reasoning for Conversational Interruptions

1

Modeling Generation Axes: Generation Model, Judge Model, Temperature, Reasoning Trace Levels (Global, Per-Turn, Per-Burst), …
Sociodemographic Generation Axes: Age, Race, Gender, Gender Presentation, …
Narrative Generation Axis: Plot Shape (Vonnegut)

2

Isolate a specific interruption for judgement

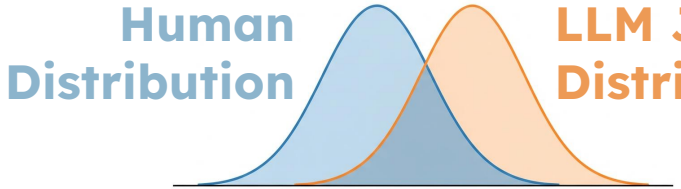
A: The committee report is basically ready to—

B: [=Right, but we haven't looked at the ↑ budget numbers—=]

A: [=—and we can file it by ↓ Monday=] if you just let me finish.

B: Oh, go ahead. (0.4) I'm ::: listening ::: ↑ really.

3

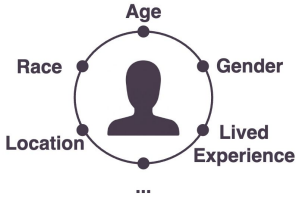


+ compare free-form reasoning traces

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Outreach

Invited Talks

Academic: George Mason University, King's College London (upcoming), Texas Tech University (Vet Med)

Industry: JLL and IPT Global in Houston, Revionics in Austin

Students: Club of Aggie Female Engineers, Sikorski Horizon (upcoming)



→ *A Cross-Community Agenda for Speech AI* out of CHI '26 workshop; 23 authors spanning 18 institutions

→ Working with INTERSPEECH to open a new track for cross-community speech AI work

Teaching and Mentoring

CSCE 676: Data Mining & Analysis *(Graduate, 125 students)*

Course Developer, TA – helped redesign the course for the AI era: e.g., students use coding agents, then validate the agent's output vs. established libraries, students practice answering interview questions

Guest Lecture – SpeechLLMs

CSCE 670: Information Storage & Retrieval *(Graduate, 125 students)*

Guest Lectures – IR Evaluation and Learning to Rank

CSCE 289: GenAI for Engineers *(Undergraduate, 30 students)*

Course Developer – course builds from foundational concepts (via hands-on API and pipeline work) to critical evaluation of model outputs

Professional Development Mastery Certificate in Instruction & Assessment, CIRTL

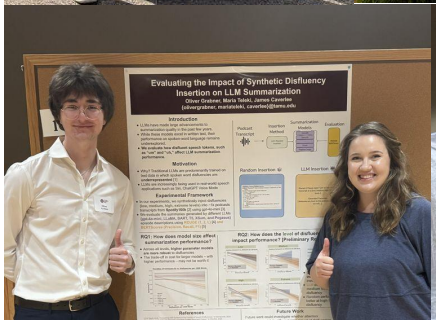
Multi-semester program in evidence-based teaching, including syllabus design, learning assessment, and inclusive pedagogy

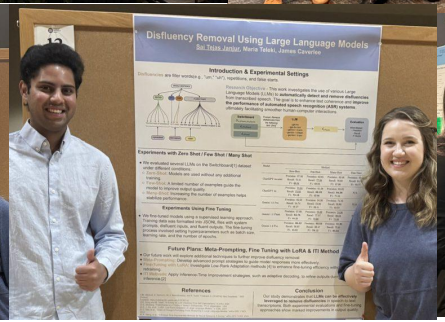
Teaching and Mentoring

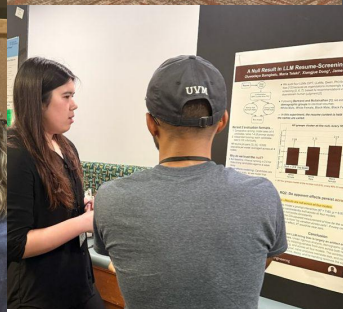












12 mentees (4 MS, 8 UG)

4 first-author papers
→ CHI ‘26 Poster
→ 2 x ICWSM ‘26 Workshop Papers
→ LREC-COLING ‘24

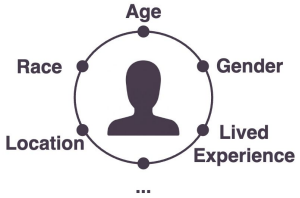
Internships & jobs at NVIDIA, Amazon, Meta, Google, Samsung, Target

Multiple students applying to MS and PhD programs – 1 currently at Cornell Tech MS

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